

LECTURE 11 LINEAR PROCESSES III: ASYMPTOTIC RESULTS

(Phillips and Solo (1992) and Phillips' Lecture Notes on Stationary and Nonstationary Time Series)

In this lecture, we discuss the LLN and CLT for a linear process $\{X_t\}$ generated as

$$\begin{aligned} X_t &= \sum_{j=0}^{\infty} c_j \varepsilon_{t-j} \\ &= C(L) \varepsilon_t, \end{aligned} \tag{1}$$

where

$$C(L) = \sum_{j=0}^{\infty} c_j L^j,$$

and $\{\varepsilon_t\}$ is a sequence of iid random variables with zero mean and finite variance. The LLN and CLT for $\{X_t\}$ rely only on the LLN and CLT for iid sequences and a certain decomposition of the lag polynomial $C(L)$.

The method also works for more general sequences with ε_t 's being independent but not identically distributed (inid) and martingale difference sequences (mds).

Definition 1 *Let $\{\mathcal{F}_t\}$ be an increasing sequence of σ -fields. Then $\{(u_t, \mathcal{F}_t)\}$ is a martingale if $E[u_t | \mathcal{F}_{t-1}] = u_{t-1}$ with probability one. The sequence $\{(\varepsilon_t, \mathcal{F}_t)\}$ is said to be a martingale difference sequence if $E[\varepsilon_t | \mathcal{F}_{t-1}] = 0$ with probability one.*

For $\{u_t\}$, the σ -field $\mathcal{F}_t$ is often taken to be $\sigma(u_t, u_{t-1}, \dots)$. Suppose that $\{(u_t, \mathcal{F}_t)\}$ is a martingale. Then $\{(u_t - u_{t-1}, \mathcal{F}_t)\}$ is an mds, since $u_t - u_{t-1} = u_t - E[u_t | \mathcal{F}_{t-1}]$, and, therefore, $E[u_t - u_{t-1} | \mathcal{F}_{t-1}] = 0$. Note that each of the requirements for $\{\varepsilon_t\}$, iid, inid, or mds, is stronger than white noise alone.

Lemma 1 (SLLN for inid sequences, White (2001), Corollary 3.9) *Let $\{\varepsilon_t\}$ be a sequence of independent random variables such that $\sup_t E[|\varepsilon_t|^{1+\delta}] < \infty$ for some $\delta > 0$. Then, $n^{-1} \sum_{t=1}^n \varepsilon_t - n^{-1} \sum_{t=1}^n E[\varepsilon_t] \rightarrow_{a.s.} 0$.*

Lemma 2 (SLLN for mds, White (2001), Exercise 3.77) *Let $\{(\varepsilon_t, \mathcal{F}_t)\}$ be an mds such that $\sup_t E[|\varepsilon_t|^{2+\delta}] < \infty$ for some $\delta > 0$. Then, $n^{-1} \sum_{t=1}^n \varepsilon_t \rightarrow_{a.s.} 0$.*

Lemma 3 (CLT for inid sequences, White (2001), Theorem 5.10) *Let $\{\varepsilon_t\}$ be a sequence of independent random variables such that $E[\varepsilon_t] = 0$ for all t , $\sup_t E[|\varepsilon_t|^{2+\delta}] < \infty$ for some $\delta > 0$, and for all n sufficiently large $n^{-1} \sum_{t=1}^n E[\varepsilon_t^2] > \delta_0 > 0$. Then, $n^{-1/2} \sum_{t=1}^n \varepsilon_t / (n^{-1} \sum_{t=1}^n E[\varepsilon_t^2])^{1/2} \rightarrow_d N(0, 1)$.*

Lemma 4 (CLT for mds, White (2001), Corollary 5.26) *Let $\{(\varepsilon_t, \mathcal{F}_t)\}$ be a mds such that $\sup_t E[|\varepsilon_t|^{2+\delta}] < \infty$ for some $\delta > 0$. Suppose that for all n sufficiently large $n^{-1} \sum_{t=1}^n E[\varepsilon_t^2] > \delta_0 > 0$, and $n^{-1} \sum_{t=1}^n \varepsilon_t^2 - n^{-1} \sum_{t=1}^n E[\varepsilon_t^2] \rightarrow_p 0$. Then, $n^{-1/2} \sum_{t=1}^n \varepsilon_t / (n^{-1} \sum_{t=1}^n E[\varepsilon_t^2])^{1/2} \rightarrow_d N(0, 1)$.*

Lemma 5 (CLT for strictly stationary and ergodic mds) *Let $\{(\varepsilon_t, \mathcal{F}_t)\}$ be a strictly stationary and ergodic mds such that $E[\varepsilon_t^2] < \infty$. Then $n^{-1/2} \sum_{t=1}^n \varepsilon_t \rightarrow_d N(0, E[\varepsilon_t^2])$.*

Beveridge and Nelson (BN) decomposition

First, we discuss an algebraic decomposition of a lag polynomial into long-run and transitory elements. The decomposition was introduced by Beveridge and Nelson (1981).

Lemma 6 *Let $C(L) = \sum_{j=0}^{\infty} c_j L^j$. Then*

(a) $C(L) = C(1) - (1-L)\tilde{C}(L)$, where $\tilde{C}(L) = \sum_{j=0}^{\infty} \tilde{c}_j L^j$ with $\tilde{c}_j = \sum_{h=j+1}^{\infty} c_h$.

(b) If $\sum_{j=1}^{\infty} j^{1/2} |c_j| < \infty$, then $\sum_{j=0}^{\infty} \tilde{c}_j^2 < \infty$.

(c) If $\sum_{j=1}^{\infty} j |c_j| < \infty$, then $\sum_{j=0}^{\infty} |\tilde{c}_j| < \infty$.

Proof. For part (a), write

$$\begin{aligned} \sum_{j=0}^{\infty} c_j L^j &= \sum_{j=0}^{\infty} c_j - \sum_{j=1}^{\infty} c_j \\ &+ \left(\sum_{j=1}^{\infty} c_j - \sum_{j=2}^{\infty} c_j \right) L \\ &+ \left(\sum_{j=2}^{\infty} c_j - \sum_{j=3}^{\infty} c_j \right) L^2 \\ &+ \dots \\ &+ \left(\sum_{j=h}^{\infty} c_j - \sum_{j=h+1}^{\infty} c_j \right) L^h \\ &+ \dots \end{aligned}$$

Rearranging the terms,

$$\begin{aligned} \sum_{j=0}^{\infty} c_j L^j &= \sum_{j=0}^{\infty} c_j \\ &- (1-L) \sum_{j=1}^{\infty} c_j \\ &- (1-L) \sum_{j=2}^{\infty} c_j L \\ &- \dots \\ &- (1-L) \sum_{j=h+1}^{\infty} c_j L^h \\ &- \dots \\ &= \sum_{j=0}^{\infty} c_j - (1-L) \sum_{j=0}^{\infty} \left(\sum_{h=j+1}^{\infty} c_h \right) L^j \\ &= C(1) - (1-L)\tilde{C}(L). \end{aligned}$$

For part (b),

$$\begin{aligned}
\sum_{j=0}^{\infty} \tilde{c}_j^2 &= \sum_{j=0}^{\infty} \left(\sum_{h=j+1}^{\infty} c_h \right)^2 \\
&\leq \sum_{j=0}^{\infty} \left(\sum_{h=j+1}^{\infty} |c_h| \right)^2 \\
&= \sum_{j=0}^{\infty} \left(\sum_{h=j+1}^{\infty} |c_h|^{1/2} h^{1/4} |c_h|^{1/2} h^{-1/4} \right)^2 \\
&\leq \sum_{j=0}^{\infty} \left(\sum_{h=j+1}^{\infty} |c_h| h^{1/2} \right) \left(\sum_{h=j+1}^{\infty} |c_h| h^{-1/2} \right) \\
&\leq \left(\sum_{h=0}^{\infty} |c_h| h^{1/2} \right) \sum_{j=0}^{\infty} \sum_{h=j+1}^{\infty} |c_h| h^{-1/2}.
\end{aligned}$$

Next, consider $\sum_{j=0}^{\infty} \sum_{h=j+1}^{\infty} |c_h| h^{-1/2}$. The term $|c_1|$ appears in the sum only once, when $j = 0$. The term $|c_2|$ appears in the sum twice, when $j = 0, 1$. Hence, $|c_h|$ appears when $j = 0, 1, \dots, h-1$, total h times. Therefore,

$$\begin{aligned}
\sum_{j=0}^{\infty} \sum_{h=j+1}^{\infty} |c_h| h^{-1/2} &= \sum_{j=0}^{\infty} |c_j| j^{-1/2} j \\
&= \sum_{j=0}^{\infty} |c_j| j^{1/2},
\end{aligned}$$

and

$$\sum_{j=0}^{\infty} \tilde{c}_j^2 \leq \left(\sum_{j=0}^{\infty} |c_j| j^{1/2} \right)^2.$$

For part (c),

$$\begin{aligned}
\sum_{j=0}^{\infty} |\tilde{c}_j| &= \sum_{j=0}^{\infty} \left| \sum_{h=j+1}^{\infty} c_h \right| \\
&\leq \sum_{j=0}^{\infty} \sum_{h=j+1}^{\infty} |c_h| \\
&= \sum_{j=0}^{\infty} |c_j| j.
\end{aligned}$$

■

Notice that the assumptions $\sum_{j=1}^{\infty} j^{1/2} |c_j| < \infty$ and $\sum_{j=1}^{\infty} j |c_j| < \infty$ are stronger than finiteness of the long-run variance $\sum_{j=1}^{\infty} |c_j| < \infty$.

According to the BN decomposition, if $\{X_t\}$ is a linear process, then

$$\begin{aligned}
X_t &= C(L) \varepsilon_t \\
&= C(1) \varepsilon_t - (1-L) \tilde{C}(L) \varepsilon_t \\
&= C(1) \varepsilon_t - (\tilde{\varepsilon}_t - \tilde{\varepsilon}_{t-1}),
\end{aligned} \tag{2}$$

where

$$\tilde{\varepsilon}_t = \tilde{C}(L) \varepsilon_t.$$

Furthermore, $\tilde{\varepsilon}_t$ has finite variance provided that $\sum_{j=0}^{\infty} j^{1/2} |c_j| < \infty$. The first summand on the right-hand side of (2), $C(1) \varepsilon_t$, is the long-run component, and $\tilde{\varepsilon}_t - \tilde{\varepsilon}_{t-1}$ is transient.

The similar decomposition exists in the vector case. The transient component has finite variance if $\sum_{j=1}^{\infty} j^{1/2} \|C_j\| < \infty$:

$$\begin{aligned} \sum_{j=0}^{\infty} \|\tilde{C}_j\|^2 &= \sum_{j=0}^{\infty} \left\| \sum_{h=j+1}^{\infty} C_h \right\|^2 \\ &\leq \sum_{j=0}^{\infty} \left(\sum_{h=j+1}^{\infty} \|C_h\| \right)^2 \\ &= \sum_{j=0}^{\infty} \left(\sum_{h=j+1}^{\infty} \|C_h\|^{1/2} h^{1/4} \|C_h\|^{1/2} h^{-1/4} \right)^2 \\ &\leq \left(\sum_{j=0}^{\infty} j^{1/2} \|C_j\| \right)^2. \end{aligned}$$

The condition in part (c) of Lemma 6 becomes $\sum_{j=1}^{\infty} j \|C_j\| < \infty$ in the vector case.

WLLN

Suppose that

- $X_t = C(L) \varepsilon_t$.
- $\{\varepsilon_t\}$ is a sequence of iid random variables with $E[|\varepsilon_t|] < \infty$ and $E[\varepsilon_t] = 0$.
- $C(L)$ satisfies $\sum_{j=1}^{\infty} j |c_j| < \infty$.

We will show that under these conditions

$$n^{-1} \sum_{t=1}^n X_t \rightarrow_p 0.$$

The key is the BN decomposition (2). Notice that $\sum_{t=1}^n (\tilde{\varepsilon}_t - \tilde{\varepsilon}_{t-1})$ is a so-called telescoping sum:

$$\begin{aligned} \sum_{t=1}^n (\tilde{\varepsilon}_t - \tilde{\varepsilon}_{t-1}) &= (\tilde{\varepsilon}_1 - \tilde{\varepsilon}_0) + (\tilde{\varepsilon}_2 - \tilde{\varepsilon}_1) + (\tilde{\varepsilon}_3 - \tilde{\varepsilon}_2) + \dots + (\tilde{\varepsilon}_n - \tilde{\varepsilon}_{n-1}) \\ &= \tilde{\varepsilon}_n - \tilde{\varepsilon}_0, \end{aligned}$$

so that

$$n^{-1} \sum_{t=1}^n X_t = C(1) n^{-1} \sum_{t=1}^n \varepsilon_t - n^{-1} (\tilde{\varepsilon}_n - \tilde{\varepsilon}_0).$$

Due to $\sum_{j=1}^{\infty} j |c_j| < \infty$, $|C(1)| < \infty$. Hence, by the WLLN for iid sequences,

$$C(1) n^{-1} \sum_{t=1}^n \varepsilon_t \rightarrow_p 0.$$

Next,

$$\Pr \left(n^{-1} |\tilde{\varepsilon}_t| > \delta \right) \leq \frac{\mathbb{E} [|\tilde{\varepsilon}_t|]}{n\delta},$$

and

$$\begin{aligned} \mathbb{E} [|\tilde{\varepsilon}_t|] &= \mathbb{E} \left[\left| \sum_{j=0}^{\infty} \tilde{c}_j \varepsilon_{t-j} \right| \right] \\ &\leq \sum_{j=0}^{\infty} |\tilde{c}_j| \mathbb{E} [|\varepsilon_{t-j}|] \\ &= \mathbb{E} [|\varepsilon_0|] \sum_{j=0}^{\infty} |\tilde{c}_j| \\ &< \infty, \end{aligned}$$

provided that $\sum_{j=1}^{\infty} j |c_j| < \infty$. Therefore,

$$n^{-1} (\tilde{\varepsilon}_n - \tilde{\varepsilon}_0) \rightarrow_p 0.$$

If we assume that $\mathbb{E} [\varepsilon_t^2] < \infty$, then we can replace $\sum_{j=1}^{\infty} j |c_j| < \infty$ with $\sum_{j=1}^{\infty} j^{1/2} |c_j| < \infty$, since

$$\Pr \left(n^{-1} |\tilde{\varepsilon}_t| > \delta \right) \leq \frac{\mathbb{E} [\tilde{\varepsilon}_t^2]}{n^2 \delta^2},$$

and $\mathbb{E} [\tilde{\varepsilon}_t^2] < \infty$ provided that $\sum_{j=1}^{\infty} j^{1/2} |c_j| < \infty$ holds.

We can prove similar WLLNs with $\{\varepsilon_t\}$ being inid or an mds by using the corresponding LLNs for inid or mds. For example, the result holds if $\{\varepsilon_t\}$ is inid, $\sup_t \mathbb{E} [|\varepsilon_t|^{1+\delta}] < \infty$ for some $\delta > 0$, and $\sum_{j=1}^{\infty} j |c_j| < \infty$.

If $\{(\varepsilon_t, \mathcal{F}_t)\}$ is an mds, the result holds with $\sup_t \mathbb{E} [|\varepsilon_t|^{2+\delta}] < \infty$ for some $\delta > 0$, and $\sum_{j=1}^{\infty} j^{1/2} |c_j| < \infty$.

So far we assumed that $\mathbb{E} [X_t] = 0$. One can modify the first assumption to allow $X_t = \mu + C(L) \varepsilon_t$, where μ is the mean of X_t . In this case, under the same set of conditions, we have $n^{-1} \sum_{t=1}^n X_t \rightarrow_p \mu$. For example, $AR(1)$ process with mean μ is given by $(1 - \phi L)(X_t - \mu) = \varepsilon_t$. If $|\phi| < 1$, then the sample average of X_t converges in probability to μ provided that the corresponding moment restrictions hold.

CLT

Suppose that

- $X_t = C(L) \varepsilon_t$.
- $\{\varepsilon_t\}$ is a sequence of iid random variables with $\mathbb{E} [|\varepsilon_t|^2] = \sigma^2 < \infty$ and $\mathbb{E} [\varepsilon_t] = 0$.
- $C(L)$ satisfies $\sum_{j=1}^{\infty} j^{1/2} |c_j| < \infty$.
- $C(1) \neq 0$.

The BN decomposition allows us to write

$$\begin{aligned} n^{-1/2} \sum_{t=1}^n X_t &= C(1) n^{-1/2} \sum_{t=1}^n \varepsilon_t - n^{-1/2} (\tilde{\varepsilon}_n - \tilde{\varepsilon}_0) \\ &= C(1) n^{-1/2} \sum_{t=1}^n \varepsilon_t + o_p(1) \\ &\rightarrow_d C(1) N(0, \sigma^2) \\ &= N(0, \sigma^2 C(1)^2). \end{aligned}$$

Here, convergence in distribution is by the CLT for iid random variables. The approach illustrates why in the serially correlated case the asymptotic variance depends on the long-run variance of $\{X_t\}$. Again, the approach can be extended to the case where $\{\varepsilon_t\}$ is inid or mds.

In the vector case, suppose that $\{\varepsilon_t\}$ is a sequence of iid k -vectors with $E[\varepsilon_t] = 0$, and $E[\varepsilon_t \varepsilon_t^\top] = \Sigma$, a finite matrix. Let $X_t = C(L) \varepsilon_t$, and $\sum_{j=0}^{\infty} j^{1/2} \|C_j\| < \infty$, $C(1) \neq 0$. Since $n^{-1/2} \sum_{t=1}^n \varepsilon_t \rightarrow_d N(0, \Sigma)$, we have that

$$n^{-1/2} \sum_{t=1}^n X_t \rightarrow_d N\left(0, C(1) \Sigma C(1)^\top\right).$$

Convergence of sample variances

Estimators of coefficients in the linear regression model involve second sample moments $n^{-1} \sum_{t=1}^n X_t X_t^\top$. Here, we discuss convergence of sample second moments when $\{X_t\}$ is a linear process. We assume that $\{X_t\}$ is a scalar linear process satisfying the same assumptions as in the previous section. Write

$$\begin{aligned} X_t^2 &= (C(L) \varepsilon_t)^2 \\ &= \left(\sum_{j=0}^{\infty} c_j \varepsilon_{t-j} \right)^2 \\ &= \sum_{j=0}^{\infty} \sum_{l=0}^{\infty} c_j c_l \varepsilon_{t-j} \varepsilon_{t-l} \\ &= \sum_{j=0}^{\infty} c_j^2 \varepsilon_{t-j}^2 + 2 \sum_{j=0}^{\infty} \sum_{l>j}^{\infty} c_j c_l \varepsilon_{t-j} \varepsilon_{t-l} \\ &= \sum_{j=0}^{\infty} c_j^2 \varepsilon_{t-j}^2 + 2 \sum_{h=1}^{\infty} \sum_{j=0}^{\infty} c_j c_{j+h} \varepsilon_{t-j} \varepsilon_{t-j-h} \quad (\text{change of variable } l = j + h, \text{ so that } h = 1, 2, \dots) \\ &= B_0(L) \varepsilon_t^2 + 2 \sum_{h=1}^{\infty} B_h(L) \varepsilon_t \varepsilon_{t-h}, \end{aligned}$$

where for $h = 0, 1, \dots$,

$$\begin{aligned} B_h(L) &= \sum_{j=0}^{\infty} b_{h,j} L^j \\ &= \sum_{j=0}^{\infty} c_j c_{j+h} L^j. \end{aligned}$$

Thus,

$$\begin{aligned} B_0(L) &= \sum_{j=0}^{\infty} b_{0,j} L^j = \sum_{j=0}^{\infty} c_j^2 L^j. \\ B_1(L) &= \sum_{j=0}^{\infty} b_{1,j} L^j = \sum_{j=0}^{\infty} c_j c_{j+1} L^j. \\ &\dots \end{aligned}$$

The BN decomposition of $B_h(L)$ is

$$B_h(L) = B_h(1) - (1-L) \tilde{B}_h(L), \quad (3)$$

where

$$\begin{aligned}\tilde{B}_h(L) &= \sum_{j=0}^{\infty} \tilde{b}_{h,j} L^j, \\ \tilde{b}_{h,j} &= \sum_{l=j+1}^{\infty} b_{h,l} \\ &= \sum_{l=j+1}^{\infty} c_l c_{l+h}.\end{aligned}$$

The BN decomposition of $B_h(L)$ is valid provided that $\sum_{j=1}^{\infty} j^{1/2} |c_j| < \infty$:

$$\begin{aligned}\sum_{j=0}^{\infty} \tilde{b}_{h,j}^2 &= \sum_{j=0}^{\infty} \left(\sum_{l=j+1}^{\infty} c_l c_{l+h} \right)^2 \\ &= \sum_{j=0}^{\infty} \left(\sum_{l=j+1}^{\infty} l^{1/4} c_l c_{l+h} l^{-1/4} \right)^2 \\ &\leq \sum_{j=0}^{\infty} \left(\sum_{l=j+1}^{\infty} l^{1/2} c_l^2 \right) \left(\sum_{l=j+1}^{\infty} c_{l+h}^2 l^{-1/2} \right) \\ &\leq \left(\sum_{l=1}^{\infty} l^{1/2} c_l^2 \right) \left(\sum_{j=0}^{\infty} \sum_{l=j+1}^{\infty} c_{l+h}^2 l^{-1/2} \right) \\ &= \left(\sum_{l=1}^{\infty} l^{1/2} c_l^2 \right) \left(\sum_{l=0}^{\infty} c_{l+h}^2 l^{1/2} \right) \\ &\leq \left(\sum_{l=1}^{\infty} l^{1/2} c_l^2 \right)^2,\end{aligned}$$

and $\sum_{l=1}^{\infty} l^{1/2} c_l^2$ is finite provided that $\sum_{l=1}^{\infty} l^{1/2} |c_l|$ is finite:

$$\begin{aligned}\sum_{l=1}^{\infty} l^{1/2} c_l^2 &= \sum_{l=1}^{\infty} l^{1/2} |c_l|^2 \\ &\leq \sup_j |c_j| \sum_{l=1}^{\infty} l^{1/2} |c_l| < \infty,\end{aligned}$$

where $\sup_j |c_j| < \infty$ because $\sum_{l=1}^{\infty} l^{1/2} |c_l| < \infty$ and therefore $|c_l| \rightarrow 0$ as $l \rightarrow \infty$.

Thus, we have

$$\begin{aligned}X_t^2 &= B_0(L) \varepsilon_t^2 + 2 \sum_{h=1}^{\infty} B_h(L) \varepsilon_t \varepsilon_{t-h} \\ &= B_0(1) \varepsilon_t^2 + 2 \sum_{h=1}^{\infty} B_h(1) \varepsilon_t \varepsilon_{t-h} \\ &\quad - (1-L) \left(\tilde{B}_0(L) \varepsilon_t^2 + 2 \sum_{h=1}^{\infty} \tilde{B}_h(L) \varepsilon_t \varepsilon_{t-h} \right) \\ &= \left(\sum_{j=0}^{\infty} c_j^2 \right) \varepsilon_t^2 + u_t - (1-L) \tilde{v}_t,\end{aligned}$$

where

$$\begin{aligned} u_t &= \varepsilon_t \left(2 \sum_{h=1}^{\infty} B_h(1) \varepsilon_{t-h} \right), \\ \tilde{v}_t &= \tilde{B}_0(L) \varepsilon_t^2 + 2 \sum_{h=1}^{\infty} \tilde{B}_h(L) \varepsilon_t \varepsilon_{t-h}. \end{aligned}$$

We have

$$n^{-1} \sum_{t=1}^n X_t^2 = \left(\sum_{j=0}^{\infty} c_j^2 \right) n^{-1} \sum_{t=1}^n \varepsilon_t^2 + n^{-1} \sum_{t=1}^n u_t - n^{-1} (\tilde{v}_n - \tilde{v}_0).$$

We will show next that

$$n^{-1} \sum_{t=1}^n u_t \rightarrow_{a.s.} 0.$$

Let $\mathcal{F}_t = \sigma(\varepsilon_t, \varepsilon_{t-1}, \dots)$. We have that $\{(u_t, \mathcal{F}_t)\}$ is an mds.

Lemma 7 (White (2001), Theorem 3.76) *Let $\{(u_t, \mathcal{F}_t)\}$ be an mds. If for some $r \geq 1$, $\sum_{t=1}^{\infty} \mathbb{E} [|u_t|^{2r}] / t^{1+r} < \infty$, then $n^{-1} \sum_{t=1}^n u_t \rightarrow_{a.s.} 0$.*

We will verify that $\{u_t\}$ satisfies the condition of the above lemma. Set $r = 1$. The condition is satisfied if $\sup_t \mathbb{E} [u_t^2] < \infty$, since $\sum_{t=1}^{\infty} t^{-2} < \infty$.

$$\begin{aligned} \mathbb{E} [u_t^2] &= 4\sigma^2 \mathbb{E} \left[\left(\sum_{h=1}^{\infty} B_h(1) \varepsilon_{t-h} \right)^2 \right] \\ &= 4\sigma^4 \left(\sum_{h=1}^{\infty} B_h(1)^2 \right). \end{aligned}$$

Next,

$$\begin{aligned} \sum_{h=1}^{\infty} B_h(1)^2 &= \sum_{h=1}^{\infty} \left(\sum_{j=0}^{\infty} c_j c_{j+h} \right)^2 \\ &\leq \sum_{h=1}^{\infty} \left(\sum_{j=0}^{\infty} c_j^2 \right) \sum_{j=0}^{\infty} c_{j+h}^2 \\ &= \left(\sum_{j=0}^{\infty} c_j^2 \right) \sum_{h=1}^{\infty} \sum_{j=0}^{\infty} c_{j+h}^2 \\ &= \left(\sum_{j=0}^{\infty} c_j^2 \right) \sum_{j=1}^{\infty} j c_j^2, \end{aligned}$$

and, by the same argument as on page 3 of Lecture 10, $\sum_{j=1}^{\infty} j^{1/2} |c_j| < \infty$ implies that $\sum_{j=1}^{\infty} j c_j^2 < \infty$ as well.

As before, one can show that

$$n^{-1} (\tilde{v}_n - \tilde{v}_0) \rightarrow_p 0,$$

provided that $\sum_{j=1}^{\infty} j^{1/2} |c_j| < \infty$.

Lastly, by the WLLN for iid sequences,

$$n^{-1} \sum_{t=1}^n \varepsilon_t^2 \rightarrow_p \sigma^2.$$

Therefore,

$$\begin{aligned} n^{-1} \sum_{t=1}^n X_t^2 &\rightarrow_p \sigma^2 \sum_{j=0}^{\infty} c_j^2 \\ &= \mathbb{E} [X_t^2]. \end{aligned}$$